

Assignment 3

Christian Parker
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1.

Dilation

0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	0	0
0	1	1	1	1	1	1	1	1	1	0	0
0	1	1	1	1	1	1	1	1	1	0	0
0	1	1	1	1	1	1	1	1	1	0	0
0	1	1	1	1	0	1	1	1	1	0	0
0	1	1	1	1	1	1	1	1	1	0	0
0	0	1	1	1	1	1	1	1	1	1	0
0	0	0	0	0	0	0	1	1	1	1	0
0	0	0	0	0	0	0	0	0	0	0	0

Erosion

0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	1	1	0	0	0	0	0
0	0	0	0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0

Opening

0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1	0	0	0
0	0	0	1	1	1	1	1	0	0	0
0	0	0	0	0	1	1	1	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0

Closing

0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	1	0	0	0
0	0	1	1	1	1	1	1	0	0	0
0	0	1	1	1	1	1	1	0	0	0
0	0	1	0	0	0	1	1	0	0	0
0	0	1	0	0	0	1	1	0	0	0
0	0	0	1	1	1	1	1	0	0	0
0	0	0	0	0	0	0	0	1	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0

2. First, we calculate the closing using the following structuring element (with origin in the center):

Structuring Element (SE)

1	1	1
1	1	1
1	1	1

This fills in all the gaps that a 3x3 square cannot fit into and results in this image:

Closing of I using SE (K)

0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	1	0	0	0
0	0	1	1	1	1	1	1	0	0	0
0	0	1	1	1	1	1	1	0	0	0
0	0	1	1	1	1	1	1	0	0	0
0	0	1	1	1	1	1	1	0	0	0
0	0	0	1	1	1	1	1	0	0	0
0	0	0	0	0	0	0	0	1	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0

Now we use the following structuring element (with origin in the center) to calculate the opening of image K:

Structuring Element (SE2)

1	1	0
1	0	0
0	0	0

The result of this operation is the image J. The opening removes pixels from the image wherever the structuring element cannot fit inside the image. The right angle shape of SE2 easily fits everywhere except the bottom right corner where it cuts off the two bottom-rightmost two pixels.

3. The optimal threshold will be at the intensity where the probability of assigning the pixel to object or background is equal:

$$\frac{1}{2}p_1(T) = \frac{1}{2}p_2(T)$$

$$p_1(T) = p_2(T)$$

If the PDFs are equal anywhere, it will be within the overlap at $4 \leq x \leq 6$. Therefore, we substitute $p_1(T)$ and $p_2(T)$ as:

$$-\frac{T}{18} + \frac{1}{3} = \frac{2}{25}T - \frac{8}{25}$$

And simplify:

$$\begin{aligned} \frac{6 - T}{18} &= \frac{2T - 8}{25} \\ 150 - 25T &= 36T - 144 \\ 294 &= 61T \\ T &= \frac{294}{61} \end{aligned}$$

The optimal threshold is at $T = 4.82$.

4. (a) After one iteration the cluster assignment for each point is:

Cluster Assignment (1 iteration)

Point	Cluster Number
(-3, 4)	1
(-2, 1)	3
(-2, -4)	2
(0, -3)	2
(0, 0)	3
(2, 2)	3
(3, 1)	3
(4, 2)	3

And the cluster centers are at:

Cluster Centers (1 iteration)

Cluster Number	Center
1	(-3, 4)
2	(-1, -3.5)
3	(1.4, 1.2)

After convergence, the cluster assignments will be:

Cluster Assignment (Convergence)

Point	Cluster Number
(-3, 4)	1
(-2, 1)	1
(-2, -4)	2
(0, -3)	2
(0, 0)	3
(2, 2)	3
(3, 1)	3
(4, 2)	3

And the cluster centers are at:

Cluster Centers (Convergence)

Cluster Number	Center
1	(-2.5, 2.5)
2	(-1, -3.5)
3	(2.25, 1.25)

(b) After convergence, the cluster assignments will be:

Cluster Assignment (Convergence)

Point	Cluster Number
(-3, 4)	1
(-2, 1)	1
(-2, -4)	2
(0, -3)	2
(0, 0)	2
(2, 2)	3
(3, 1)	3
(4, 2)	3

And the cluster centers are at:

Cluster Centers (Convergence)

Cluster Number	Center
1	(-2.5, 2.5)
2	(-0.67, -2.33)
3	(3, 1.67)

- (c) The positions of the initial cluster centers can affect the number of iterations before convergence, the final cluster assignments, and the locations of the final cluster centers. The initial cluster centers from (a) took 3 iterations compared to (b)'s 2 iterations. The point at (0,0) ends up in cluster 3 in (a), but cluster 2 in (b). This coincides with the different final cluster centers we see for clusters 2 and 3. If we assume that the initial centers for clusters 1 and 2 are fixed, then the final assignment for (0,0) appears to be based on whether the initial position for cluster 3 is closer to the pair of points with negative y-values or the triplet with positive x and y-values.
- (d)

Membership Assignments for (-3,4)

Cluster Number	Assignment
1	1
2	0
3	0

Membership Assignments for (0,0)

Cluster Number	Assignment
1	.159
2	.442
3	.398